

Name: _____

Double Award Science – Chemistry Unit 2 Foundation Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /60	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	G	F	E	D	C	Max UMS
2017	10	13	16	19	23	27
2018	9	13	17	21	26	31
2019	9	14	19	24	29	34
2022	8	11	14	17	21	26
2023	13	17	21	25	29	33



Surname	Centre Number	Candidate Number
Other Names		0



GCSE – **NEW**

3430U50-1



S18-3430U50-1

SCIENCE (Double Award)

Unit 5 – CHEMISTRY 2
FOUNDATION TIER

THURSDAY, 17 MAY 2018 – MORNING

1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	5	
2.	8	
3.	5	
4.	13	
5.	8	
6.	6	
7.	11	
8.	4	
Total	60	

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.
The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.



MAY183430U50101

Answer all questions.

1. (a) Salts can be produced by reacting acids with alkalis.

(i) Complete the following equation for the reaction between an acid and alkali. [1]



(ii) **Circle** the word which best describes the reaction between an acid and an alkali. [1]

displacement

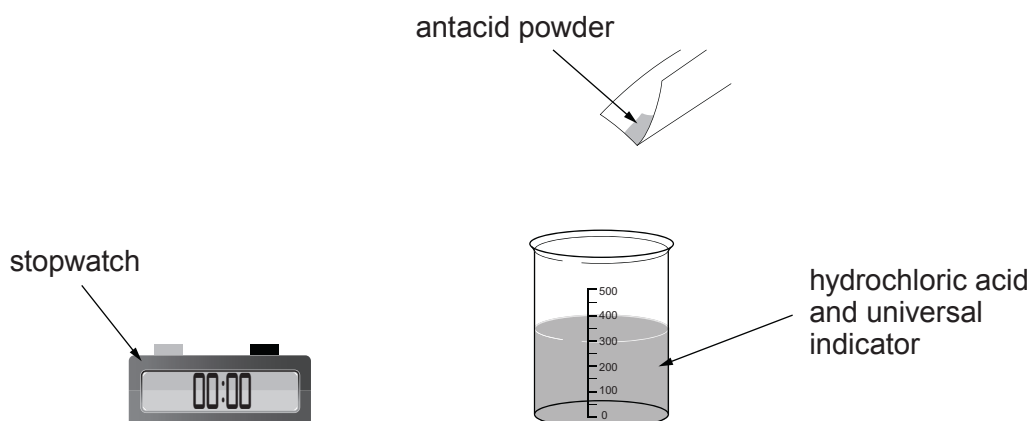
neutralisation

oxidation

reduction

(b) Indigestion can be caused by excess hydrochloric acid in the stomach. To treat indigestion, antacid powders are commonly used.

A group of pupils used the following apparatus to compare three brands of antacid powder, to see which was the most effective at treating acid indigestion.



They added an equal mass of each of the antacid powders to separate beakers, containing equal amounts of hydrochloric acid and universal indicator.

They stirred the mixture and recorded the time taken for the universal indicator to turn green in each beaker. They carried out the test three times for each antacid powder. Their results are shown in the table.



Antacid powder	Time taken for the universal indicator to turn green (min : s)			
	Result 1	Result 2	Result 3	Mean
Brand 1	5 : 25	5 : 36	5 : 14	5 : 25
Brand 2	4 : 28	3 : 20	4 : 32	4 : 30
Brand 3	2 : 28	2 : 30	2 : 44	2 : 34

- (i) State which **two** results were used to calculate the mean value for brand 2. [1]

..... and

- (ii) Convert the mean time for brand 2 into **seconds**. [1]

Mean time = s

- (iii) Give the reason why the results suggest that brand 3 is the best powder for treating acid indigestion. [1]

.....
.....

3430U501
03



2. (a) Chromium is one of the metals found in stainless steel. The equation shows how chromium is produced industrially by reacting chromium oxide with aluminium.



- (i) The reaction is highly exothermic.

Give the meaning of the term *exothermic*.

[1]

- (ii) During the reaction, oxidation and reduction happens.

I. Name the substance which is oxidised.

[1]

II. State what is meant by *reduction*.

[1]

- (iii) State what the equation tells you about the relative reactivities of chromium and aluminium.

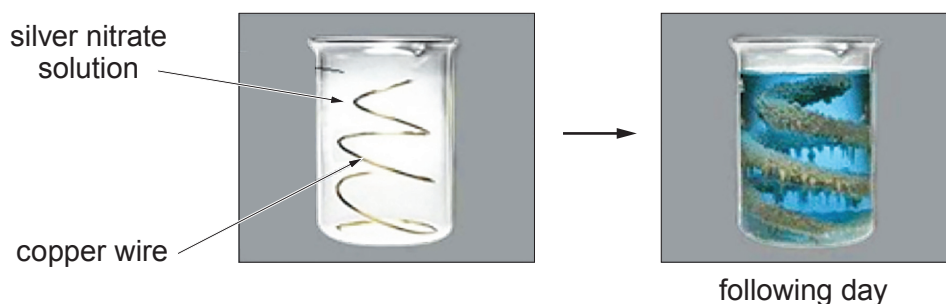
[1]



- (b) Copper is able to displace silver from a solution of silver nitrate. The equation for this reaction is given below.



A teacher demonstrated this reaction to her class. The photographs show the beaker before and after the reaction had taken place.



- (i) Explain how the changes show that this chemical reaction has taken place. [2]

.....

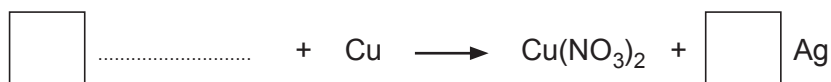
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- (ii) Complete the symbol equation for the reaction by

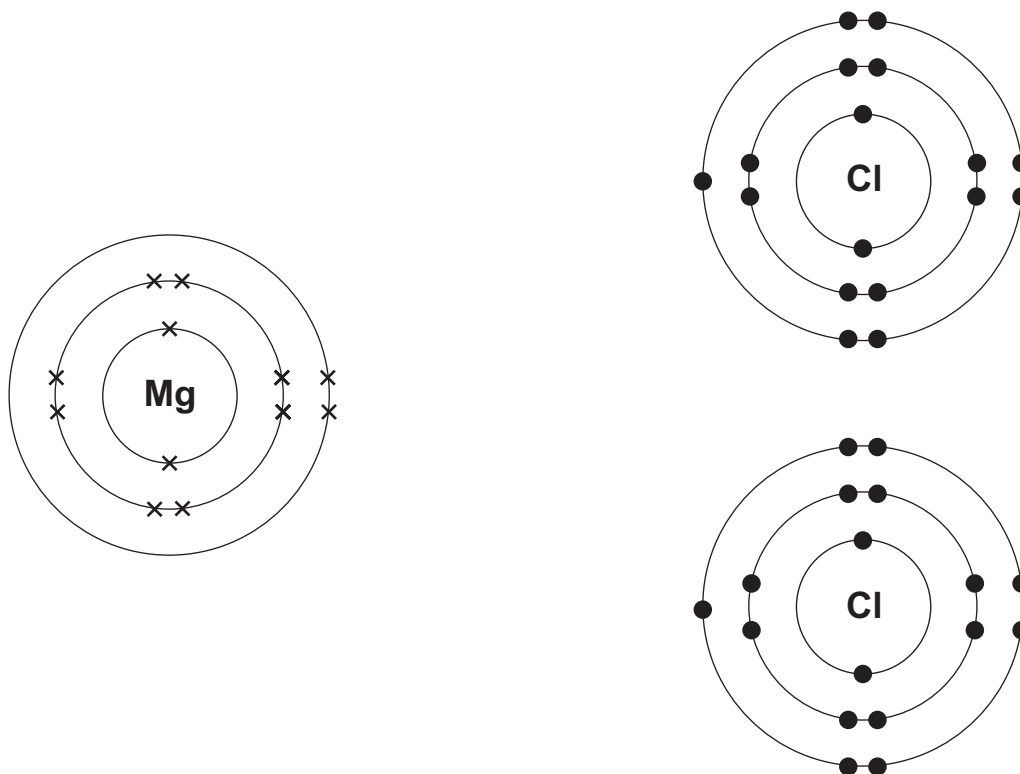
- giving the formula for silver nitrate
- balancing the overall equation

[2]



3. (a) Magnesium reacts with chlorine to form magnesium chloride.

The following diagram shows the electronic structures of magnesium and chlorine atoms.



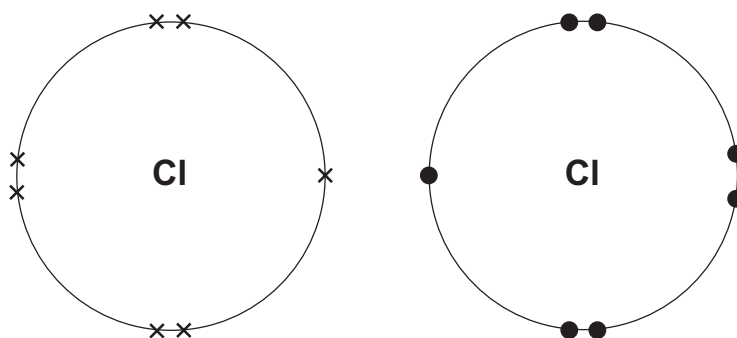
- (i) **Draw arrows on the diagram** to show how electrons are transferred between the magnesium and chlorine atoms during the formation of magnesium chloride. [1]
- (ii) **Complete the table** giving information about the charge and electronic structure of the magnesium and chloride **ions** that are formed. [2]

Ion	Charge	Electronic structure
magnesium	+2	
chloride		2,8,8



(b) Chlorine gas, Cl_2 , consists of two chlorine atoms bonded together.

- (i) Draw a diagram in the box to show how the atoms bond to form a chlorine molecule. [1]



- (ii) Give the name of this type of bonding.

.....

[1]

5

3430U501
07



4. (a) Crude oil is a fossil fuel and is described as a non-renewable resource.

(i) Describe how crude oil was formed.

[2]

.....

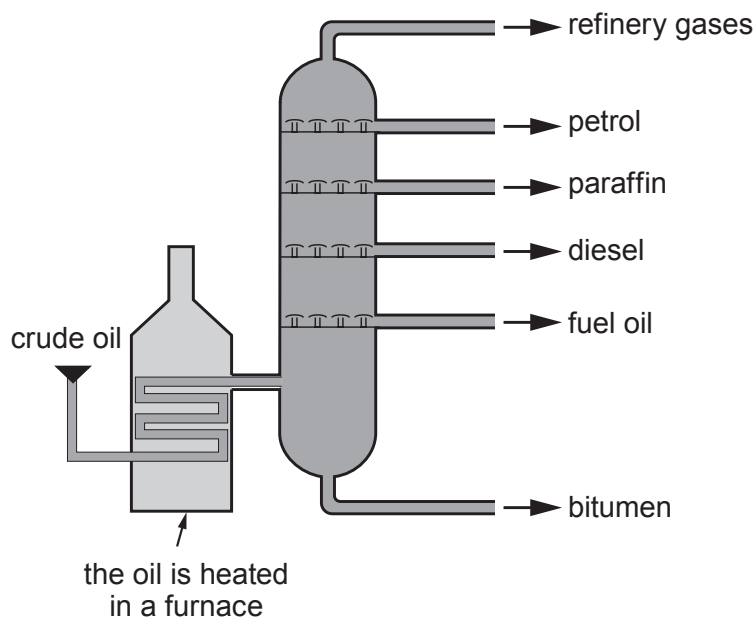
.....

(ii) Give the meaning of the term *non-renewable*.

[1]

.....

(b) To make crude oil more useful, it is separated into fractions.



(i) Complete the following sentences.

[2]

Crude oil is separated into different fractions by a process called fractional

.....

The fractions can be separated because they have different

.....

(ii) **Circle** the word which best describes crude oil.

[1]

mixture

element

compound



- (c) One of the fractions obtained from crude oil contains hexane, C_6H_{14} .

Calculate the percentage by mass of carbon in hexane.

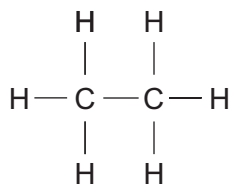
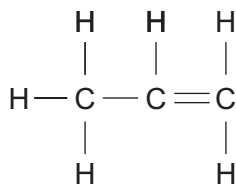
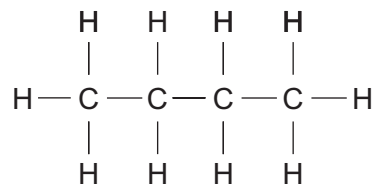
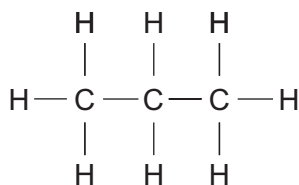
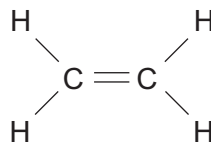
[2]

$$A_r(H) = 1 \quad A_r(C) = 12$$

Percentage = %



(d) The diagram shows the structures of five different hydrocarbons, **A-E**.

**A****B****C****D****E**

Use letters **A-E** in your answers to parts (i) and (ii).

(i) Give the structure that represents propene. [1]

(ii) Identify the structures that fit the following descriptions. [2]

Hydrocarbons with the general formula $\text{C}_n\text{H}_{2n+2}$

Unsaturated molecules



- (e) Plastics are made from chemicals that are obtained from crude oil. Supermarkets in Wales were the first in the UK to charge their customers for plastic bags. This was to reduce the amount of plastic waste generated.

Give **two** methods of plastic waste disposal that lead to environmental problems. Explain the problem linked to each method. [2]

Method 1

Problem

.....

Method 2

Problem

.....

13



5. There are a number of factors that should be taken into consideration when deciding what makes the 'best fuel'.

Information was collected about various factors for three fuels, **A**, **B** and **C**.

Fuel A

- Existing supplies will last around 50 years
- Releases 2.8 kJ of energy per gram of fuel burned
- Costs 0.03p per gram of fuel burned
- Burns very easily and no storage issues
- Releases carbon dioxide and water vapour when it burns

Fuel B

- There are infinite supplies of this fuel
- Releases 44.1 kJ of energy per gram of fuel burned
- Costs 0.18p per gram of fuel burned
- Burns very easily but can be difficult to store
- Releases water vapour when it burns

Fuel C

- Existing supplies will last around 250 years
- Releases 1.2 kJ of energy per gram of fuel burned
- Costs 0.04p per gram of fuel burned
- Burns very easily and fairly easy to store
- Releases carbon dioxide, sulfur dioxide and water vapour when it burns

This information was analysed by a group of students to decide what they considered to be the 'best fuel'.



- (a) Give the reason why the information about how easily each fuel burns was not useful to the students when reaching their decision. [1]

.....

.....

- (b) One of the students based his decision purely on a judgement of the available supply of each fuel. Choose the order that shows his conclusion. Place a tick (✓) in the appropriate box. [1]

best fuel	A	A	B	B	C	C
	B	C	A	C	A	B
	C	B	C	A	B	A
	☐	☐	☐	☐	☐	☐

- (c) Which of the following statements best describes how the fuels affect the environment when they burn? Tick (✓) the correct answer and give the reason for your choice. [2]

all of the fuels contribute to acid rain and global warming when they burn

☐

fuels **A** and **C** contribute to acid rain and global warming when they burn

☐

only fuel **C** contributes to acid rain and global warming when it burns

☐

none of the fuels contribute to acid rain and global warming when they burn

☐

Reason

.....



(d) The cost efficiency of fuel **A** can be calculated as follows:

cost efficiency = $\frac{2.8}{0.03}$ = 93.3 kJ/p

Use the information given for fuel **B** and this example to calculate the cost efficiency of fuel **B**. [2]

Cost efficiency = kJ/p

(e) The students eventually agreed on the following rank order for the fuels.



In the table below, tick (✓) **all** the statements that are **correct** and could therefore have been used in deciding upon this order. [2]

(✓)

fuel C will run out after fuels A and B	
fuel C is easier to store than fuel A	
fuel A burns more easily than fuel C	
fuel B is the cleanest fuel	
fuel B is easier to store than fuel C	
fuel B will never run out	
fuel A is less harmful to the environment than fuel C	
fuel A is less cost efficient than fuel B	



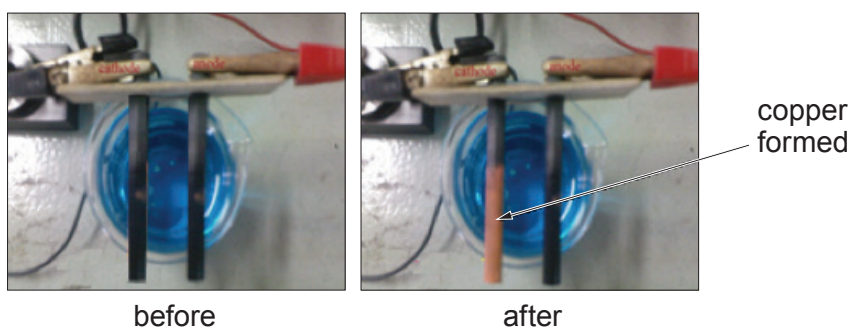
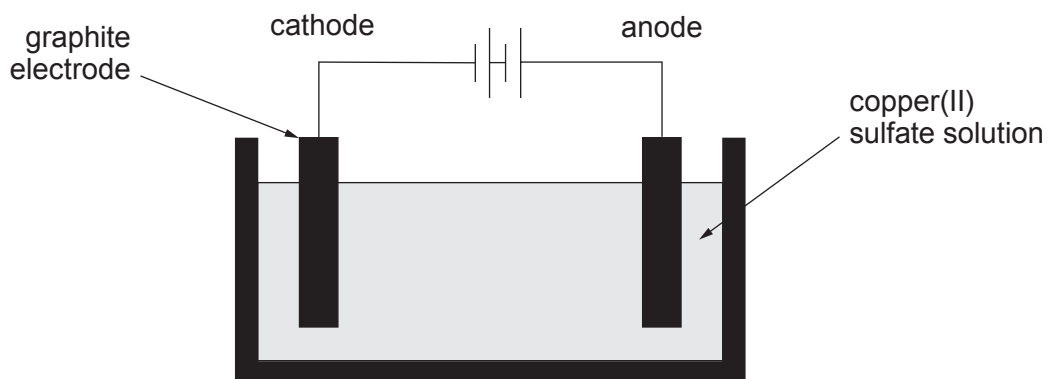
Examiner
only

-
- The diagram illustrates the three stages of a simple distillation process:
- Stage 1:** A beaker containing a mixture with a thermometer.
 - Stage 2:** A funnel-shaped condenser is placed over the beaker.
 - Stage 3:** The condensed liquid is collected in a separate container.

[6 QER]

7. A group of students carried out an investigation into the electrolysis of copper(II) sulfate solution. They used the apparatus shown to test the hypothesis:

“the mass of copper that forms on the cathode increases as the time increases”



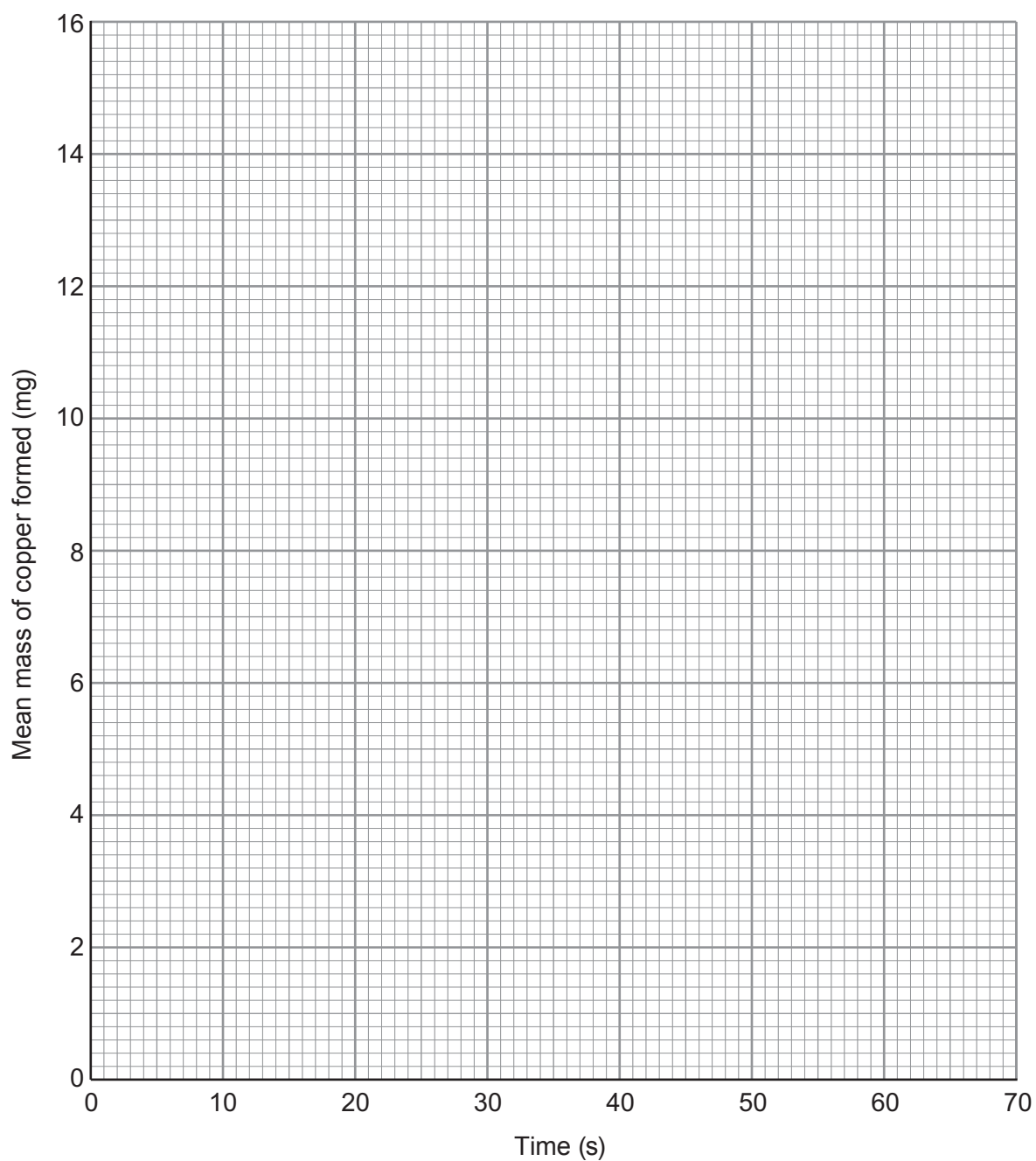
To test the hypothesis, they weighed the cathode before placing it into the copper(II) sulfate solution and then again after allowing electrolysis to take place for varying times.

Their results are shown below.

Time (s)	Mass of copper formed (mg)		
	1	2	Mean
0	0	0	0
10	2.8	3.2	3.0
20	4.8	5.0	4.9
30	8.2	7.8	8.0
40	10.8	11.2	11.0
50	12.9	13.1	13.0
60	15.8	16.0	15.9



- (a) On the grid below, plot the mean mass of copper formed against time. Draw a suitable line. [3]



- (b) (i) Use the results collected at 30s and the following equation to calculate the percentage variation in these measurements. [2]

$$\text{percentage variation} = \frac{\text{furthest mass from the mean} - \text{mean mass}}{\text{mean mass}} \times 100$$

Percentage variation = %

- (ii) The mass of copper formed is lower than expected. Give the most likely reason for this difference. [1]

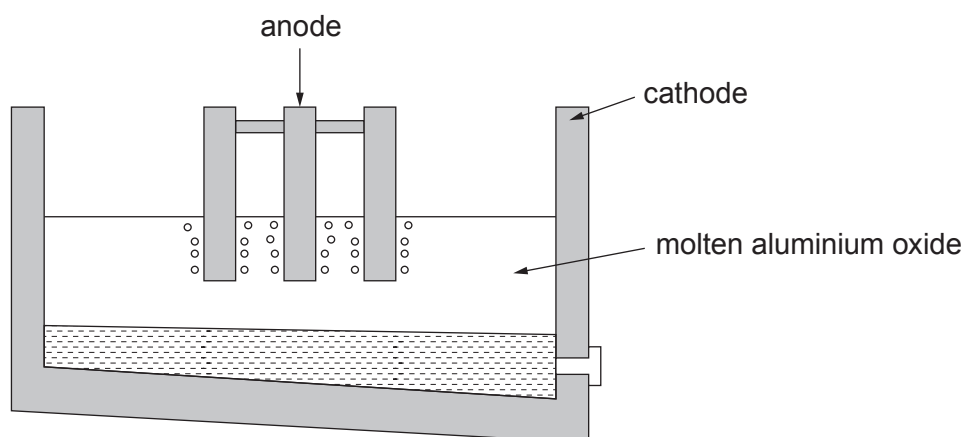
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- (c) (i) Aluminium is extracted from molten aluminium oxide by electrolysis.



- I. Explain why aluminium forms at the cathode.

[2]

.....

.....

.....

- II. Complete and balance the equation for the overall reaction that takes place.

[2]



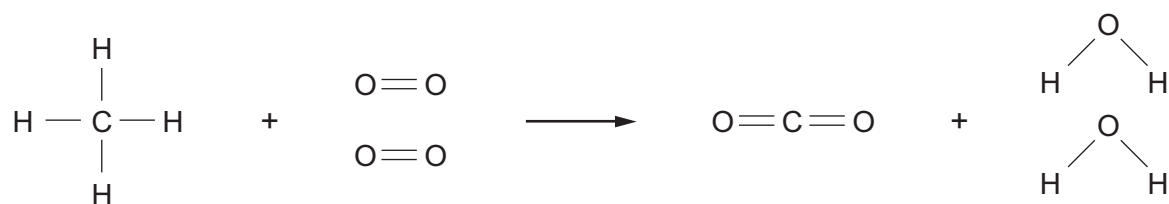
- (ii) Potassium can also be extracted through electrolysis of potassium carbonate.

Write the **formula** of potassium carbonate to complete the equation for the overall reaction.

[1]



8. The burning of methane in air can be represented by the following equation.



The bond energies are given in the table below.

Bond	Bond energy (kJ)
C — H	413
O = O	498
O — H	464
C = O	805

- (a) Use the bond energy values to calculate the energy released when **all** the bonds in the carbon dioxide and water molecules are formed. [2]

Energy released = kJ



- (b) The energy needed to break **all** the bonds in the methane and oxygen molecules is 2648 kJ.

Calculate the overall energy change for this reaction and use this value to explain why the reaction is exothermic. [2]

Overall energy change = kJ

.....
.....

4

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		



THE PERIODIC TABLE

0

7

6

5

4

3

Group

2

1

24



1
H
Hydrogen
1

7 Li Lithium 3	9 Be Beryllium 4
23 Na Sodium 11	24 Mg Magnesium 12
39 K Potassium 19	40 Ca Calcium 20
86 Rb Rubidium 37	88 Sr Strontium 38
133 Cs Caesium 55	137 Ba Barium 56
223 Fr Francium 87	226 Ra Radium 88
	227 Ac Actinium 89

11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86

24

45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30
89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48
139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
Other Names		0

**GCSE**

3430U50-1



S19-3430U50-1

SCIENCE (Double Award)**Unit 5 – CHEMISTRY 2
FOUNDATION TIER****THURSDAY, 16 MAY 2019 – MORNING****1 hour 15 minutes**

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	6	
2.	14	
3.	6	
4.	6	
5.	13	
6.	15	
Total	60	

3430U501
01**ADDITIONAL MATERIALS**

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

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Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **5(c)** is a quality of extended response (QER) question where your writing skills will be assessed.

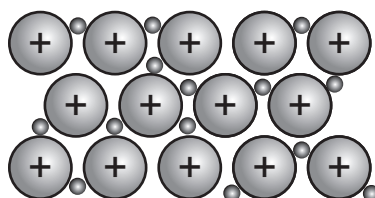
The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.



MAY193430U50101

Answer all questions.

1. The following diagram can be used to represent the structure of silver.



- (a) Select the correct words from the box to complete the sentences that follow. [3]

electrons	tightly	protons
malleable	neutrons	brittle
ductile	irregularly	loosely

Silver is a solid because the metal ions are packed.

Silver is a good conductor of electricity because the are delocalised and free to move.

Silver dents when hammered because it is

- (b) Silver can also exist as nano-sized particles. These are used in plasters, socks and deodorant sprays.

- (i) **Underline** the correct word in the brackets to complete each sentence. [2]

Nano-sized silver particles range from (**1-100 mm** / **1-100 nm** / **1-100 cm**) in size.

Nano-sized silver particles have (**the same** / **different** / **stronger**) properties compared to bulk silver.

- (ii) Give the property of nano-sized silver particles that allows them to be used in plasters, socks and deodorants. [1]

.....



2. A teacher was marking the exercise book of a student who had completed a topic of work on acids and alkalis.

- (a) The table below shows the results the student had recorded from an experiment to investigate the pH values of some common laboratory chemicals.

The teacher has circled three errors for the student to correct.

Chemical name	Chemical formula	Colour with universal indicator	pH	Acid, alkali or neutral
sulfuric acid	H_2SO_4	green	1	acid
ethanoic acid	CH_3COOH	orange	4	alkali
calcium hydroxide	$\text{Ca}(\text{OH})_2$	purple	12	alkali
water	H_2O	green	5	neutral
sodium carbonate	NaCO_3	blue	10	alkali

error 1

error 3

error 2

- (i) Correct the errors the teacher has circled.

[3]

Correction to **error 1**

Correction to **error 2**

Correction to **error 3**

- (ii) There is another error in the table that the teacher has not spotted.
Circle this error in the table.

[1]



- (b) Another piece of work in the exercise book showed the results of some tests that the student had carried out to identify some of the ions in the chemicals being investigated. However the table had not been fully completed by the student.

Ion being identified	Test	Result
carbonate CO_3^{2-}	add hydrochloric acid	bubbles formed as gas is produced
sulfate SO_4^{2-}	add solution	white precipitate is formed

- (i) Use the names of chemicals from the box below to complete the table. [2]

silver nitrate	hydrogen	sodium hydroxide
	barium chloride	oxygen
universal indicator	carbon dioxide	chlorine

- (ii) Describe the test that can be used to identify the gas produced when hydrochloric acid is added to a compound containing carbonate ions. [1]

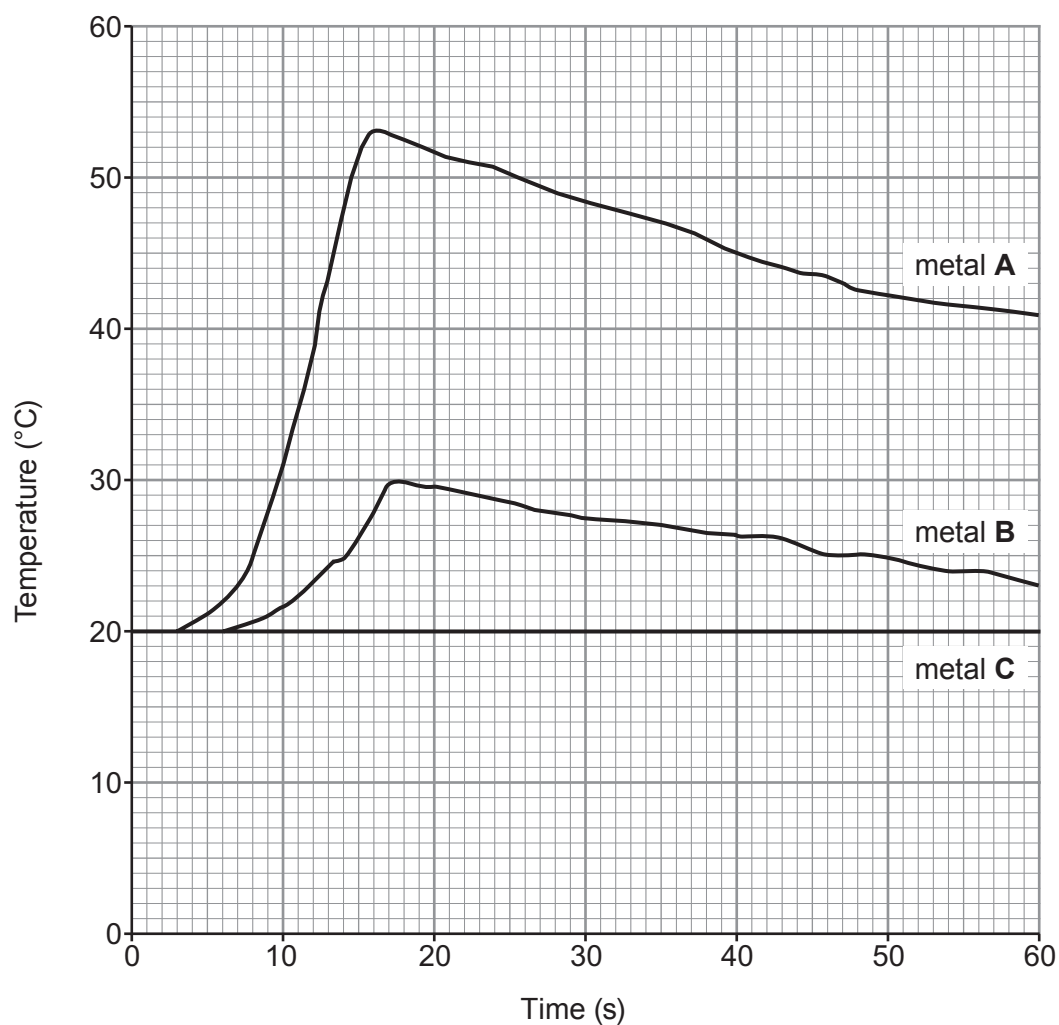
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- (c) The final piece of work marked by the teacher was an experiment that had been completed to investigate the temperature change when different metals react with hydrochloric acid.

The graphs of the results obtained by the student are shown below.



- (i) Use the graphs to give

I. the **letter** of the metal that did not react with hydrochloric acid. [1]

.....

II. the maximum temperature **rise** for metal **A**. [1]

..... °C

- (ii) State the term used to describe a reaction that gives out heat. [1]

.....



(d) When metals react with acids they form a salt and hydrogen gas.

- (i) Complete the word equation for the reaction between magnesium and sulfuric acid. [1]

magnesium + sulfuric acid $\longrightarrow$ + hydrogen

- (ii) Give the test that can be used to identify hydrogen gas. Include the expected observation. [1]

.....
.....

(iii) The salt formed when zinc reacts with hydrochloric acid is zinc chloride.

- I. Give the formulae of the ions present in zinc chloride. [1]

..... and

- II. Give the formula of zinc chloride. [1]

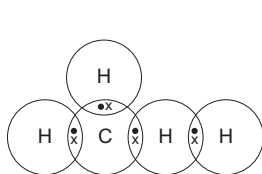
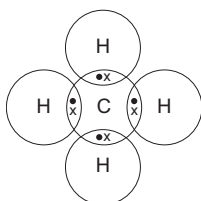
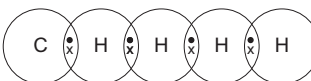
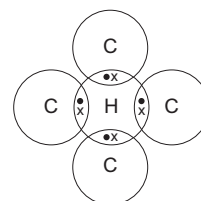
.....



3. (a) Methane gas, CH_4 , contains the elements carbon and hydrogen.

Element	Electronic structure
carbon	2,4
hydrogen	1

- (i) I. Put a **tick** (✓) in the box below the diagram which shows how the atoms are bonded in one molecule of methane. [1]


☐

☐

☐

☐

- II. **Circle** the name given to the type of bonding found in methane. [1]

covalent

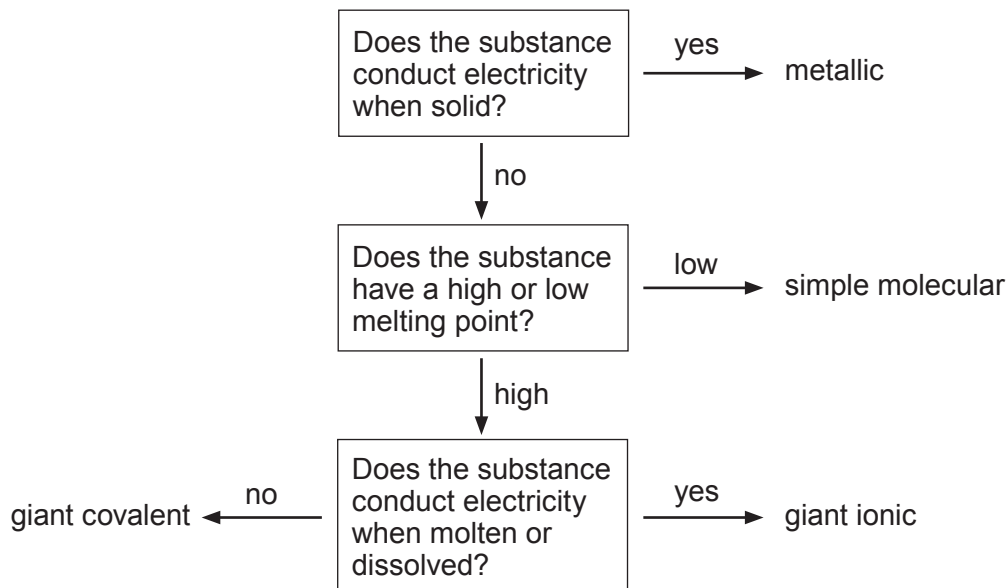
ionic

giant covalent

metallic



- (ii) The flow chart below can be used to identify the type of structure found in different substances.



- I. Use the flow chart to complete the table.

[1]

Substance	Melting point (°C)	Electrical conductivity	Type of structure
A	2072	conducts only when molten	
B	−182	does not conduct electricity	
C	1610	does not conduct electricity	

- II. Give the **letter** of the substance, **A**, **B** or **C**, that is most likely to be methane. [1]

Substance



- (b) Butene, C_4H_8 , also contains the elements carbon and hydrogen. Calculate the percentage by mass of hydrogen in butene. [2]

$$A_r(H) = 1 \quad A_r(C) = 12$$

Percentage = %

Examiner
only

6



4. An alloy is a material composed of a mixture of elements, at least one of which is a metal. The table lists the composition and common uses of different alloys containing silver and gold.

Name of alloy	Composition of alloy by mass (%)	Uses of alloy
amalgam	mercury 48 %, silver 25 %, tin 15 %, copper 12 %	dental fillings, mining
green gold	gold 75 %, silver 6-24 %, copper %	Nobel Prize medals, decoration
nordic gold	gold 89 %, aluminium 5 %, zinc 5 %, tin 1 %	coins, decoration
solder	tin 90 %, silver 5 %, copper 5 %	joining electrical components
sterling silver	silver 92.5 %, platinum 4 %, germanium 1.5-3 %, zinc 0.5-2 %	decoration, plumbing, instruments, jewellery
white gold	gold 75 %, palladium 10 %, nickel 10 %, zinc 5 %	decoration, jewellery

(a) **Circle** the number of alloys that contain an element from Group 4 of the Periodic Table. [1]

0 1 2 3 4 5 6

(b) **Circle** the correct percentage range for the mass of copper that can be found in green gold. [1]

0-25 % 1-19 % 19-25 % 6-24 %



(c) Tick (✓) the statement that **best** describes the composition of the alloys listed. [1]

the alloys all contain at least one metal

☐

the alloys all contain at least two metals

☐

the alloys all contain at least three metals

☐

the alloys all contain at least four metals

☐

(d) Tick (✓) the statement that **best** describes the decorative uses of the alloys listed. [1]

all of the alloys are used for decorative purposes

☐

all of the alloys containing gold are used for decorative purposes

☐

all of the alloys containing silver are used for decorative purposes

☐

none of the alloys containing silver are used for decorative purposes

☐

(e) A solder joint in an electrical circuit contains 0.00011 kg of silver.

Use this information and the composition of solder given in the table to calculate the mass of tin in the solder joint. [2]

Mass = kg

6



5. (a) Metal ores are the materials found in the Earth's crust from which metals can be extracted. The following table gives information about some common metal ores.

Name of metal ore	Metal extracted from ore	Formula of ore
anglesite	lead	PbCl_2
copper pyrite	copper	CuFeS_2
cryolite	aluminium	Na_3AlF_6
salt petre	potassium	KNO_3
syberite	gold	AgAuTe_2
tin pyrite	tin	$\text{Cu}_2\text{FeSnS}_4$

Use the information in the table to give

- (i) the **total** number of atoms shown in the formula of cryolite, [1]

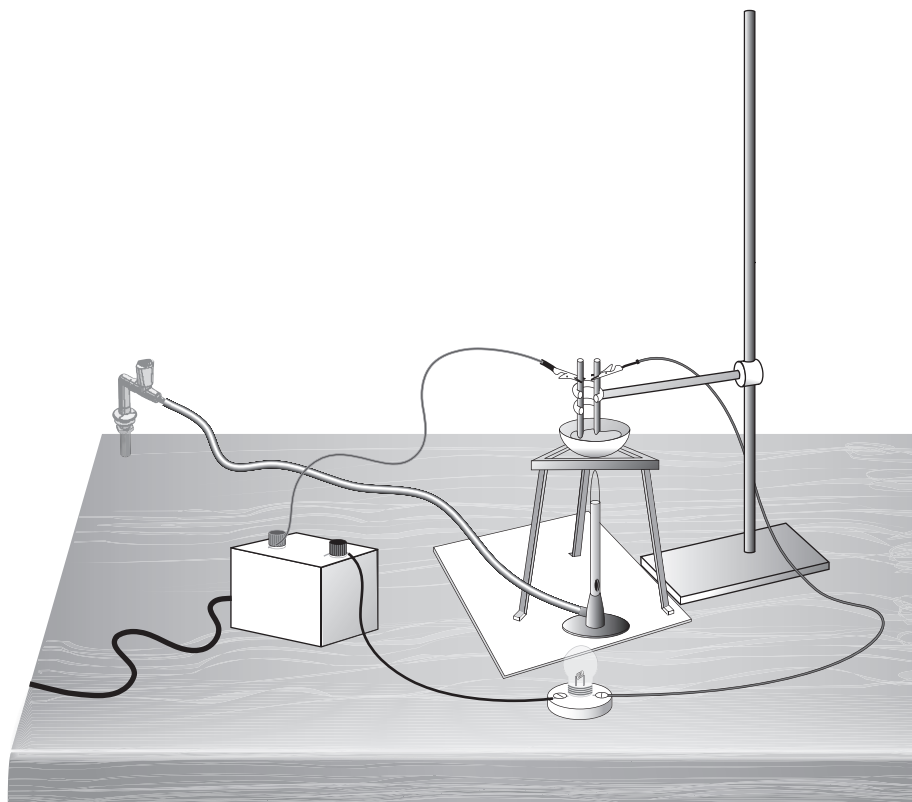
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- (ii) the name of the ore that contains a ratio of 1:4 metal to non-metal atoms. [1]

.....



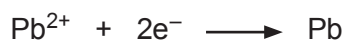
- (b) The following diagram shows how the electrolysis of anglesite (lead chloride) can be carried out in the laboratory.



- (i) Give the reason why lead chloride needs to be melted for electrolysis to take place. [1]

- (ii) Explain why the chloride ions move towards the positive electrode. [2]

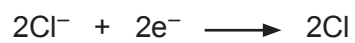
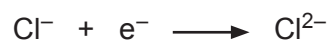
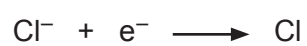
- (iii) The process occurring at the negative electrode is shown by the following equation.



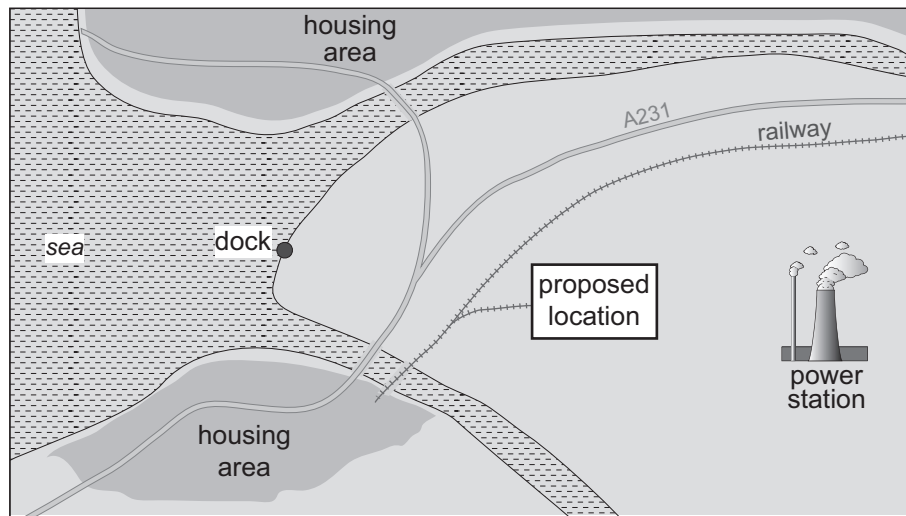
Use this **equation** to state what is meant by reduction. [1]



- (iv) Tick (✓) the equation that shows the reaction taking place at the positive electrode. [1]

☐☐☐☐☐

(c) The following diagram shows a proposed location for a new aluminium works in Wales.



Use the information in the diagram to explain why this would make a suitable location for an aluminium works. [6 QER]

.....

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6. Crude oil can be separated into simpler mixtures, called fractions, which contain hydrocarbon compounds with boiling points in a similar range.

(a) The table lists the properties of some fractions obtained from crude oil.

Fraction	Number of carbon atoms in fraction	Boiling point range (°C)	Colour of fraction	Flame when burning	Ease of burning
fuel oil	1-4	-170 to 20	colourless	clean	very easy
petrol	5-10	20 to 70	pale yellow	clean	easy
naphtha	8-12	70 to 120	yellow	some soot	quite easy
kerosene	10-16	120 to 240	dark yellow	smoky	quite difficult
diesel oil	15-30	240 to 350	brown	smoky	difficult

Use the information in the table to describe how the burning of the fractions depends upon the size of the molecules. [2]

.....

.....

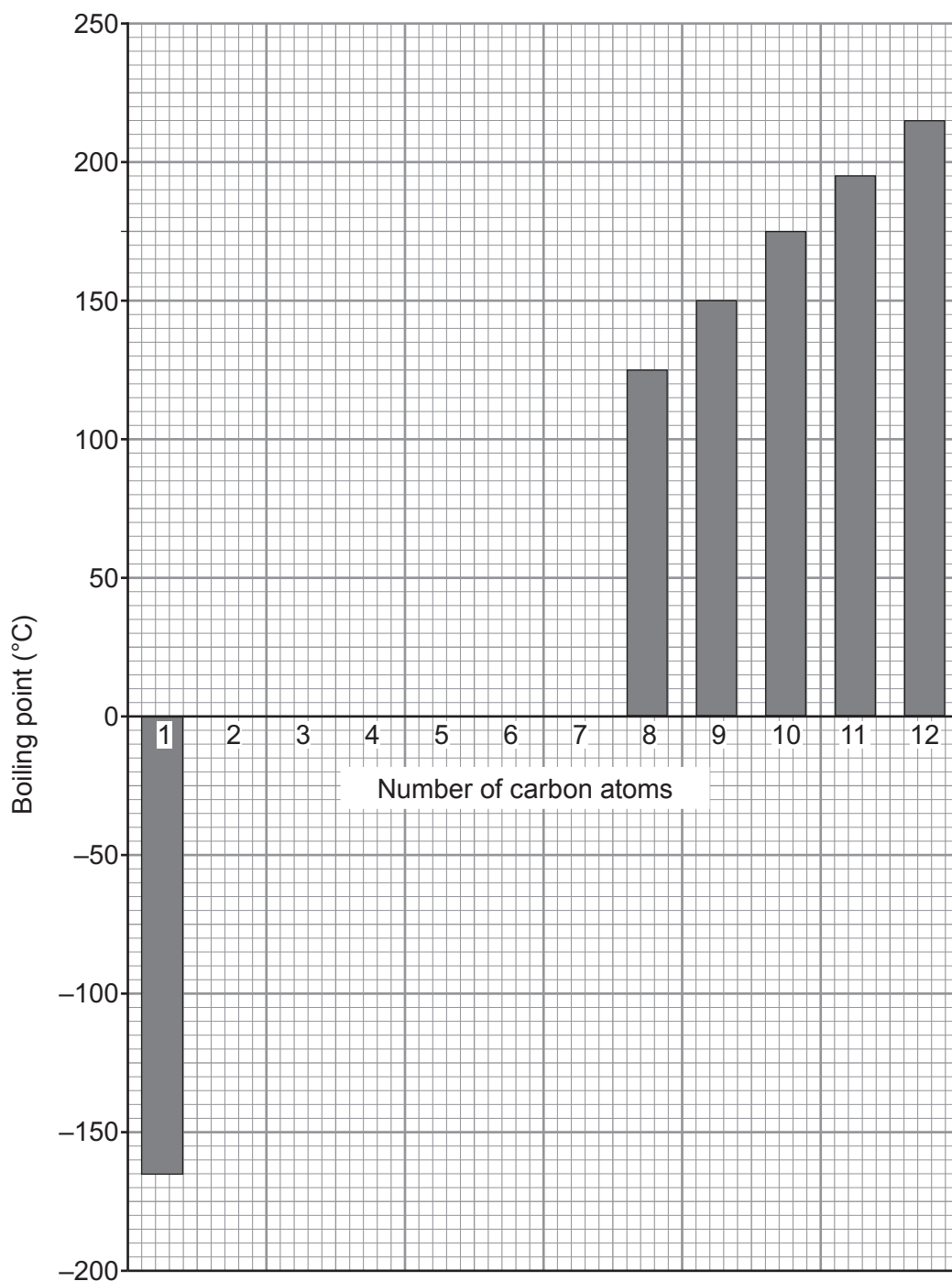
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- (b) The boiling points of hydrocarbons containing 1 to 12 carbon atoms are shown in the table below. The boiling point for the hydrocarbon with 7 carbon atoms is missing.

Number of carbon atoms	Boiling point (°C)
1	-165
2	-90
3	-40
4	10
5	35
6	70
7	
8	125
9	150
10	175
11	195
12	215



- (i) Complete the bar chart below. Some of the bars have already been drawn. [2]



- (ii) Use a ruler to draw a trend line **on the chart** and use this to estimate the boiling point of the hydrocarbon with 7 carbon atoms. [2]

Boiling point °C



(c) Many of the fractions obtained from crude oil are used as fuels.

(i) The fire triangle shows the factors necessary to start and maintain a fire.

State **one** method that could be used to safely put out a small amount of spilled petrol burning on the floor. Give the reason why your chosen method would work. [1]

Method

Reason

.....

(ii) One of the hydrocarbons in petrol is pentane, C_5H_{12} .

Complete and balance the symbol equation for the complete combustion of pentane. [2]



(iii) Hydrogen fuel cells are now used in many cars instead of petrol. The overall change inside a hydrogen fuel cell is the same as when hydrogen burns.

Explain why using hydrogen fuel cells in cars is better for the environment than petrol. [2]

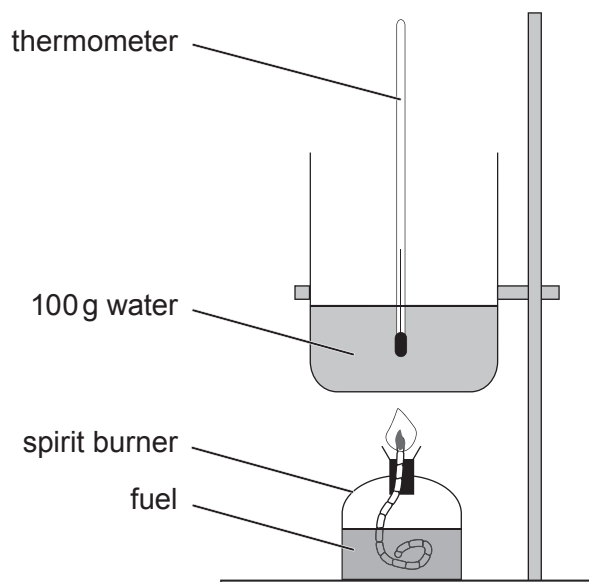
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- (d) It is possible to compare the energy released when different fuels are burned using the following apparatus.



To calculate the energy released per gram of fuel burned, the following equation is used.

$$\text{energy released per gram of fuel (J/g)} = \frac{\text{mass of water} \times 4.2 \times \text{temperature rise (}^{\circ}\text{C)}}{\text{mass of fuel used (g)}}$$

- (i) Apart from measuring the mass of water, describe **all** the measurements that would need to be taken to be able to calculate the energy released per gram of fuel burned. [2]

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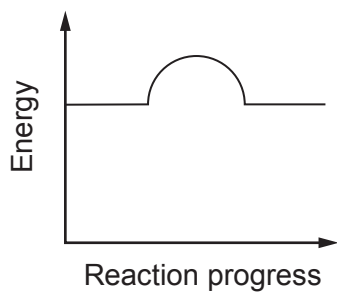
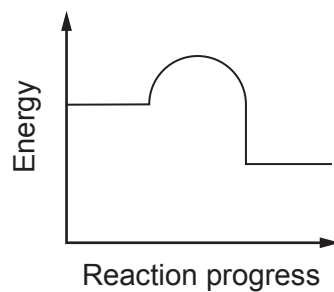
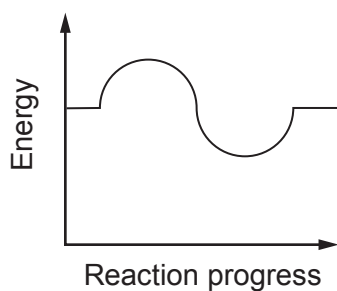
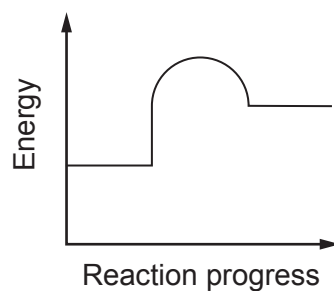
- (ii) When comparing the energy released from different fuels, 100 g of water should be used each time. State **one** other variable that should be controlled. [1]

.....

.....



(iii) Tick (✓) the energy profile diagram that represents the combustion of fuels. [1]


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END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		



THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0



1
28

1 H Hydrogen 1																				4 He Helium 2	
7	Li	9	Be	11	B	12	C	14	N	16	O	19	F	20	Ne						28
23	Na	24	Mg	27	Al	28	Si	31	P	32	S	35.5	Cl	40	Ar						
39	K	40	Ca	45	Sc	48	Ti	51	V	52	Cr	55	Mn	56	Fe	59	Co	59	Ni	63.5	Cu
86	Rb	88	Sr	89	Y	91	Zr	93	Nb	96	Mo	99	Tc	101	Ru	103	Rh	106	Pd	108	Ag
133	Cs	137	Ba	139	La	179	Hf	181	Ta	184	W	186	Re	190	Os	192	Ir	195	Pt	197	Au
223	Fr	226	Ra	227	Ac	227	Fr	226	Ra	227	Ac	227	Fr	226	Ra	227	Ac	227	Fr	226	Ra

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE**

3430U50-1



Z22-3430U50-1

FRIDAY, 27 MAY 2022 – MORNING**SCIENCE (Double Award)****Unit 5 – CHEMISTRY 2****FOUNDATION TIER**

1 hour 15 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	13	
2.	9	
3.	9	
4.	8	
5.	6	
6.	7	
7.	8	
Total	60	

3430U501
01**INSTRUCTIONS TO CANDIDATES**

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **5** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

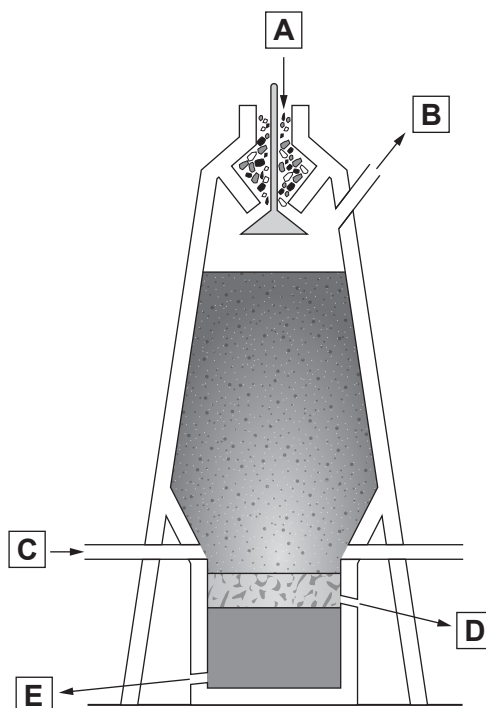


JUN223430U50101

Answer **all** questions.

1. (a) The diagram shows a blast furnace which is used to extract iron.

Labels **A**, **B**, **C**, **D** and **E** show where substances enter and leave the furnace.



- (i) Give the letter, **A**, **B**, **C**, **D** or **E**, that shows where

iron ore enters the furnace

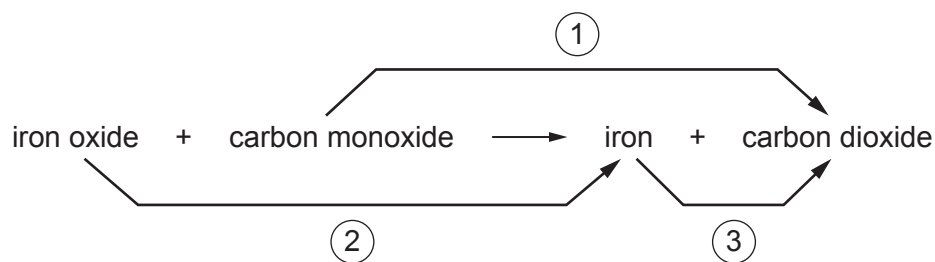
waste gases leave the furnace

slag is removed

[3]



- (ii) One of the main reactions inside the blast furnace is represented by the following equation.



Give the number of the arrow that shows oxidation.

[1]

.....

- (iii) Limestone is added to the furnace to remove impurities.

The chemical name for limestone is calcium carbonate.

Calcium carbonate contains the ions Ca^{2+} and CO_3^{2-} .

Circle the correct formula for calcium carbonate.

[1]

CaCO

Ca₂CO₃

CaCO₃

Ca(CO₃)₂

- (iv) One of the waste gases that leaves the blast furnace is carbon dioxide, CO_2 .

Put a tick (✓) in the box next to the diagram that best represents a carbon dioxide molecule.

[1]

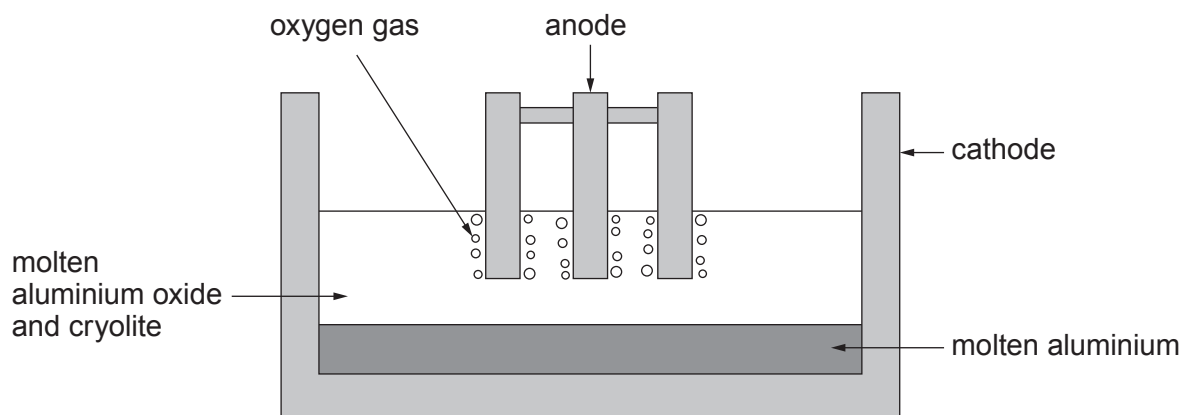

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(b) The diagram shows an electrolysis cell used in the extraction of aluminium.



(i) Draw **one** line from each term to its correct definition.

[3]

Term	Definition
anode	positive electrode
electrolyte	a substance that removes impurities
electrolysis	a substance that is split up during the process
	using electricity to make a compound
	using electricity to split up a compound
	negative electrode



- (ii) Underline the correct word in the brackets to complete each sentence. [3]

Cryolite is added to lower the (**density** / **melting point** / **boiling point**) of the electrolyte.

When choosing a location for an aluminium plant in the UK, it is important to be near a port to (**import** / **export** / **clean**) the aluminium ore.

At the temperature inside the cell, the aluminium is produced as a (**solid** / **liquid** / **gas**).

- (iii) The equation for the reaction that takes place during the extraction of aluminium is given below.



Choose a number from the box to balance the equation.

[1]

2	4	6
---	---	---

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05

13



2. (a) Hannah and Evan investigated the temperature increase when sodium hydroxide solution was added to dilute hydrochloric acid.

The method they used to collect their results is given below.

1. Measure 25 cm^3 of hydrochloric acid into a polystyrene cup.
2. Record the temperature of the hydrochloric acid.
3. Measure 25 cm^3 of sodium hydroxide and add it to the hydrochloric acid.
4. Record the highest temperature of the mixture.
5. Calculate the temperature increase for the reaction.

- (i) Choose apparatus from the box to complete the following sentences. [2]

balance	Bunsen burner	thermometer	measuring cylinder
test tube	stopwatch	tongs	beaker

Hannah and Evan used a to measure 25 cm^3 of hydrochloric acid accurately.

Hannah and Evan used a to measure the temperature of the mixture.

- (ii) Give **one** way that Hannah and Evan could check that the method produces consistent results. [1]

.....

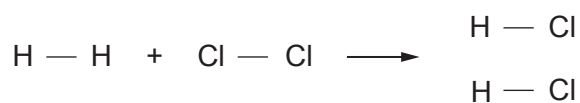
- (iii) Hannah and Evan calculated a temperature increase of 17°C .

Give the term used to describe a reaction that gives a temperature increase. [1]

.....



- (b) When chlorine reacts with hydrogen, hydrogen chloride is formed.



The bond energies are given in the table.

Bond	Bond energy (kJ)
H — H	436
Cl — Cl	243
H — Cl	432

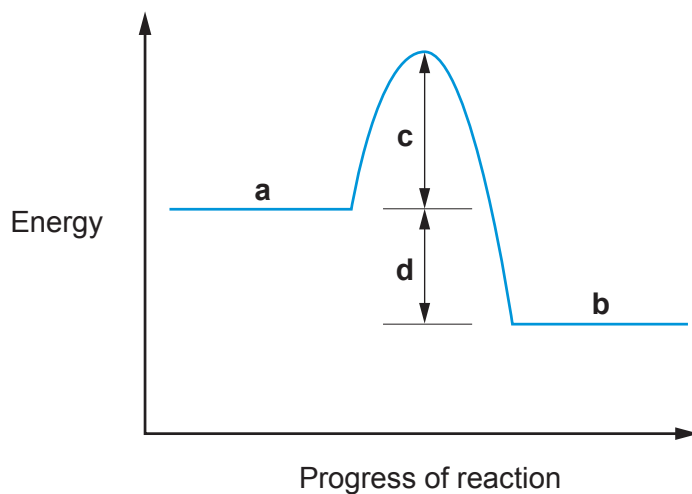
- (i) The energy needed to break the bonds in the hydrogen and chlorine molecules is 679 kJ. Show how this value is calculated. [1]

- (ii) Calculate the energy released when **two** molecules of hydrogen chloride are formed. [2]

Energy released = kJ



- (iii) The energy profile diagram for the reaction between hydrogen and chlorine is given below.



Give the letter that shows

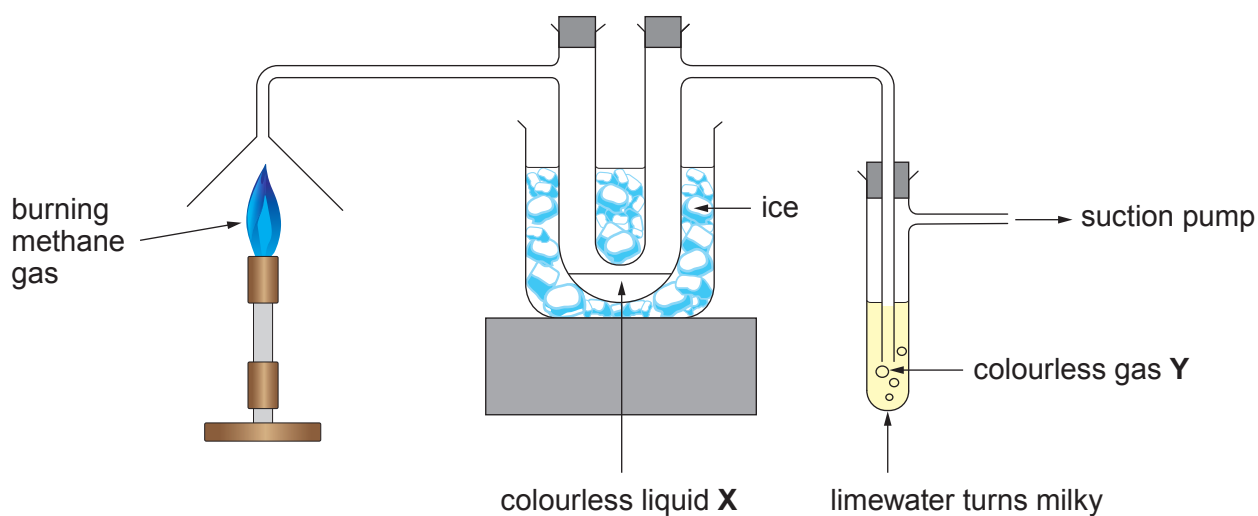
the activation energy

the overall energy change for the reaction

[2]



3. (a) The diagram shows apparatus that can be used to investigate the products formed when methane gas burns. Methane has the formula CH_4 .



- (i) Name the gas present in the air that is needed for methane to burn. [1]

.....

- (ii) Name the products formed. [2]

Colourless liquid **X**

Colourless gas **Y**

- (iii) Tick (✓) the box that gives the name of the group of substances that methane belongs to. [1]

alkenes

☐

monomers

☐

polymers

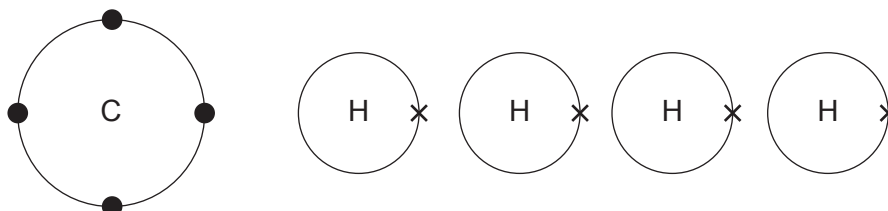
☐

alkanes

☐


- (b) Methane has the formula CH_4 . Its molecules consist of one carbon atom and four hydrogen atoms bonded together.

- (i) Draw a diagram in the box to show how the atoms bond to form a methane molecule. [2]



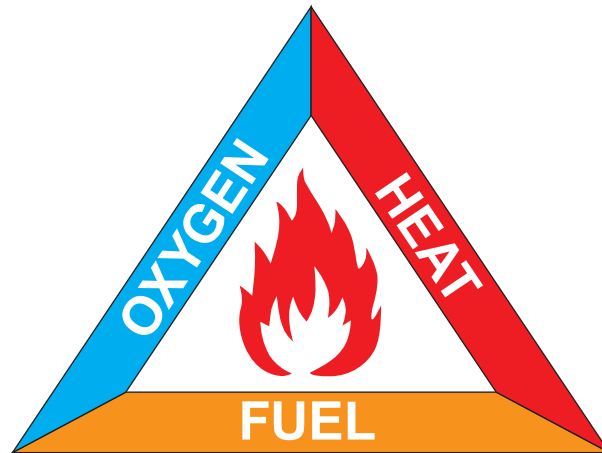
- (ii) Give the name of this type of bonding.

[1]

.....



- (c) The fire triangle can be used to explain how fires are extinguished.



The following table gives information about three different types of fire and the methods used to extinguish them.

However, each of the three fires has an **error** in **one** of the columns.

Fire	Type of fire	Firefighting method	How method works
1	chip pan fire	tea towel	removes the heat
2	bonfire	fire blanket	removes the heat
3	electrical fire	fire breaks	removes the fuel

Circle the errors for fires **1** and **2** and then correct them in the table below.

The error for fire **3** has already been circled and the correction given.

[2]

Fire	Correction
1	
2	
3	Fire breaks are used to extinguish a forest fire



4. (a) Supermarkets across the UK now charge customers a minimum of 5p for every plastic carrier bag supplied.

The reason for the charge is to reduce the number of plastic carrier bags used by consumers and therefore reduce waste and litter.

Since the introduction of the charge for plastic carrier bags in Wales, the number of these bags used by consumers has fallen by over 70 %.

Similarly, reports in England claim that supermarkets have issued 83 % fewer bags since the charge was introduced.

It is estimated that every person in the UK currently uses around 25 plastic carrier bags per year, compared to around 140 before charges were introduced.

The majority of the money retailers generate from sales of carrier bags is donated to good causes. A survey of retailers across England and Wales reported that £87 million had been donated to good causes since the introduction of the 5p charge, amounting to 4p for every bag sold.



- (i) Use the information in the passage to decide whether the following statements are **true** or **false**.

Put a tick (✓) in the correct column for each statement.

[3]

Statement	True	False
The number of plastic bags used in Wales and England has reduced since charging for them		
Retailers donate all the money generated from the sale of plastic bags to good causes		
Plastic bags are no longer used		
The charge for plastic bags has totally stopped their use in Wales		
The use of plastic bags leads to environmental problems		
The charge for plastic bags is beneficial to good causes		

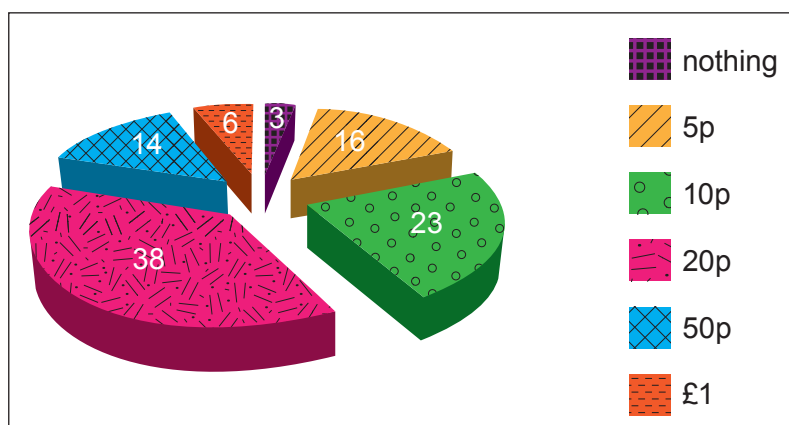


- (ii) How many fewer plastic bags will each person in the UK use over a 5-year period, based on the estimated number of bags used per person before and after the charge was introduced? [2]

Number of bags =

- (b) More than half of all consumers still regularly buy plastic bags.

The pie chart shows the results of a survey where 100 consumers were asked what is the most they would pay for a plastic bag.



Give the number of consumers that would be prepared to pay **more than** 5p for a plastic bag. [2]

Number of consumers =

- (c) Give **one** property of plastics which leads to long-term environmental problems. [1]

.....



6. Polymer gels are commonly used in disposable nappies.

A company that manufactures disposable nappies was investigating the effect of temperature on the mass of water the polymer gel in their nappies is able to absorb.

- (a) The results collected using water at 40 °C are given below. The initial mass of the polymer gel bead was 0.035 g.

Time (hours)	Mass of bead (g)	Mass of water absorbed by bead (g) (to 1 decimal place)
0	0.035	0.0
2	4.048	4.0
4	6.030	6.0
6	7.280	7.2
8	7.891	7.9
10	8.181	8.1
12	8.181	8.1

- (i) The percentage increase in the mass of the bead is calculated using the following equation.

$$\text{percentage increase} = \frac{\text{mass of water absorbed}}{\text{initial mass of bead}} \times 100$$

Calculate the percentage increase in the mass of the bead after 2 hours.
Give your answer to the nearest whole number.

[1]

Percentage increase = %

- (ii) What property of polymer gels does the figure calculated in part (i) demonstrate?

[1]

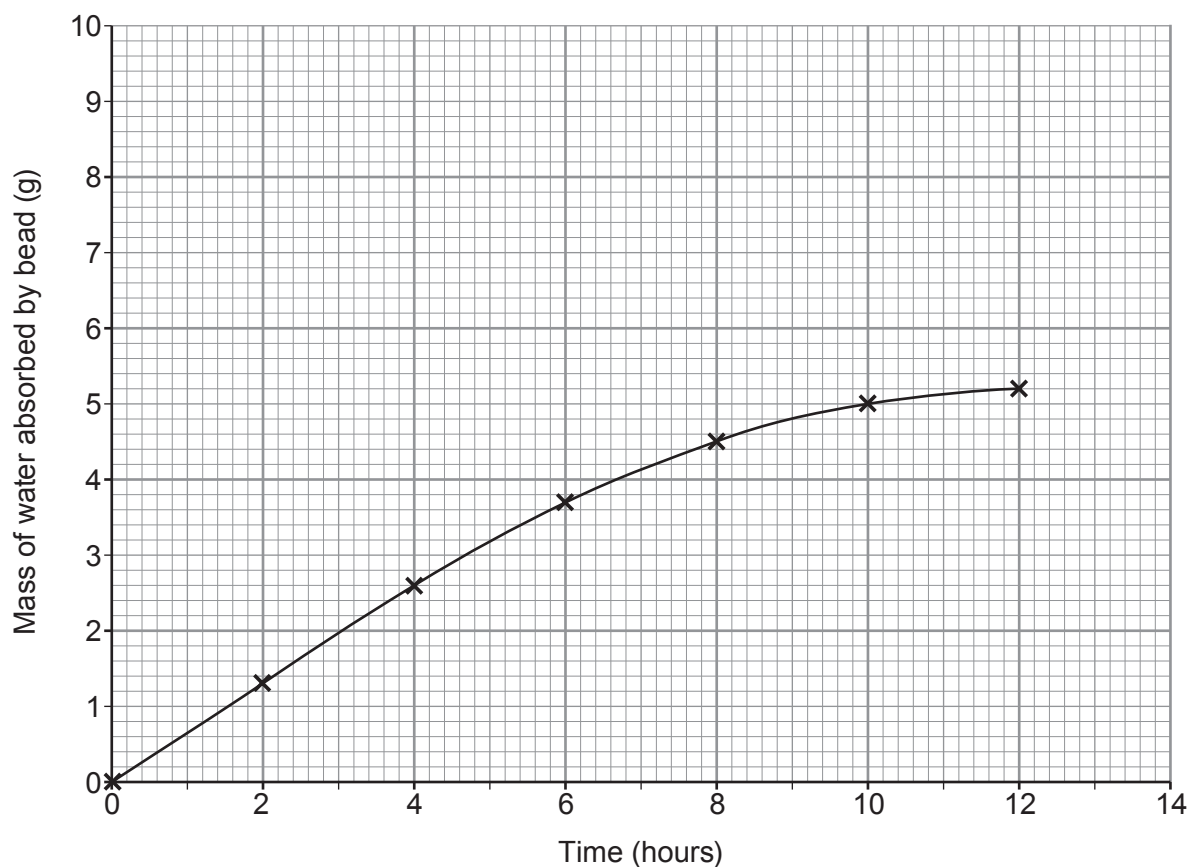
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- (b) (i) On the grid below, plot the results using water at 40 °C and draw a suitable line. Use the mass of water absorbed by the bead to 1 decimal place.

The results using water at 10 °C have already been plotted.

[3]



- (ii) Give **two** differences between the absorbing properties of the bead using water at 10 °C and at 40 °C.

[2]

Difference 1

.....

Difference 2

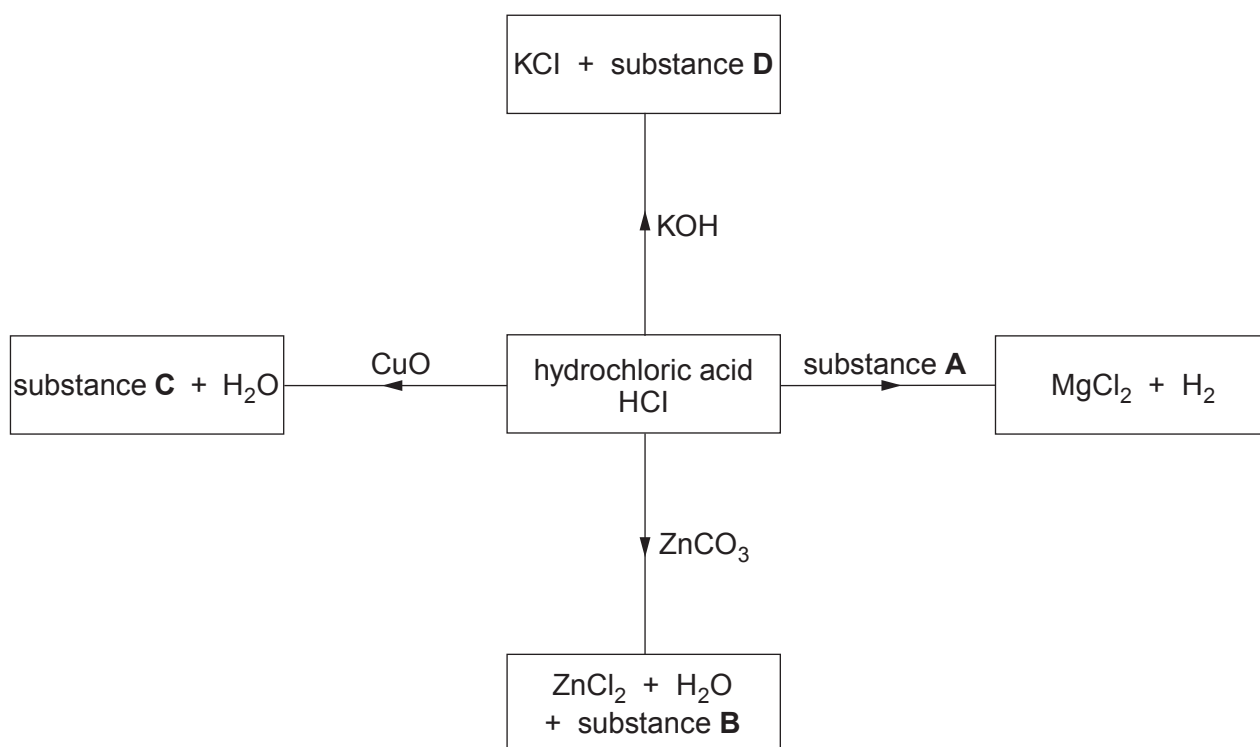
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7. The reactions of acids with metals, bases and carbonates are summarised in the following equations.

- acid + metal $\rightarrow$ salt + hydrogen
- acid + base $\rightarrow$ salt + water
- acid + carbonate $\rightarrow$ salt + water + carbon dioxide

(a) The diagram shows some reactions of hydrochloric acid, HCl.



(i) Give the **names** of substances **A** and **B**.

[2]

Substance **A**

Substance **B**

(ii) Give the **formulae** of substances **C** and **D**.

[2]

Substance **C**

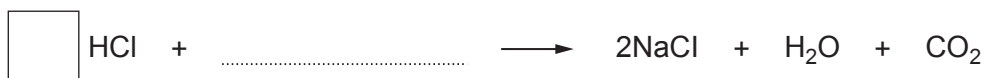
Substance **D**



- (b) Complete the equation for the reaction between hydrochloric acid and sodium carbonate by

- writing the formula of sodium carbonate on the dotted line
- putting a number into the box to balance the equation

[2]

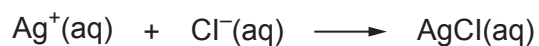
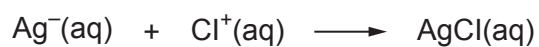
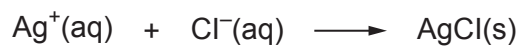
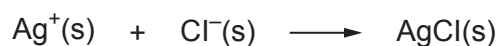
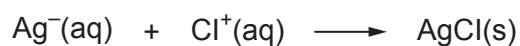


- (c) Silver nitrate solution is used to identify the chloride ions present in hydrochloric acid.

- (i) Give the observation made when silver nitrate solution is added to hydrochloric acid. [1]

.....

- (ii) Put a tick (✓) in the box next to the correct ionic equation for the reaction between silver nitrate and hydrochloric acid. [1]


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END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group

1 2

3 4 5 6 7 0

1 H Hydrogen 1															4 He Helium 2		
7 Li Lithium 3	9 Be Beryllium 4						11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10					
23 Na Sodium 11	24 Mg Magnesium 12						27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18					
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
First name(s)		0



GCSE

3430U50-1



S23-3430U50-1

MONDAY, 22 MAY 2023 – MORNING

SCIENCE (Double Award)
Unit 5 – CHEMISTRY 2
FOUNDATION TIER

1 hour 15 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	6	
3.	12	
4.	6	
5.	6	
6.	6	
7.	15	
Total	60	



JUN233430U50101

Answer **all** questions.

1. (a) (i) The box below contains the names of some common types of smart material.

photochromic pigment	hydrogel	shape memory alloy
shape memory polymer	thermochromic pigment	

Choose from the box the type of smart material that is used in:

- I. disposable nappies, [1]

.....

- II. lenses for sunglasses. [1]

.....

- (ii) What must be done to a thermochromic pigment for it to change colour? [1]

Tick (✓) the box next to the correct answer.

heat it

☐

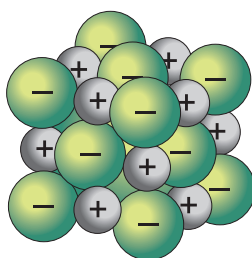
add water

☐

place it in sunlight

☐


- (b) (i) The following diagram shows a giant ionic structure.



Giant ionic structures have high melting points and conduct electricity when dissolved or molten.

Use the diagram and your knowledge to draw **one** line from each property to its correct explanation. [2]

Property

Explanation

high melting point

ions are regularly arranged

strong bonds between ions

ions are free to move

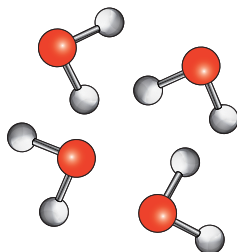
conducts electricity when dissolved or molten

ions cannot move

weak bonds between ions



(ii) The following diagram shows a simple covalent structure.



Simple covalent structures have low melting points and do not conduct electricity.

Use the diagram and your knowledge to draw **one** line from each property to its correct explanation. [2]

Property

Explanation

low melting point

does not conduct electricity

strong forces between molecules

molecules cannot move

weak forces between molecules

molecules are not charged

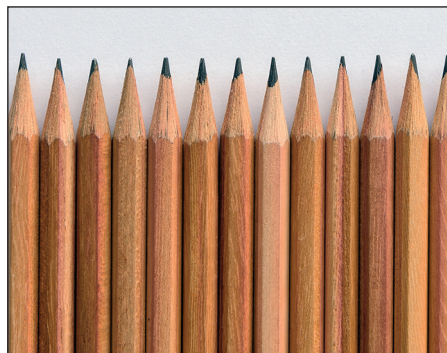
molecules are tightly packed



- (c) Diamond and graphite have giant covalent structures. Diamond is commonly used in drill tips, whereas graphite is used in pencils.



drill tips



pencils

Use your knowledge of the properties of diamond and graphite to underline the correct word in the brackets to complete each sentence. [2]

Diamond is used in drill tips because it is (**hard** / **soft** / **malleable**) and has a high melting point.

Graphite is used in pencils because the layers of atoms are held together weakly and are able to slide (**through** / **into** / **over**) each other.



2. Iron is extracted from iron oxide inside the blast furnace.



(a)

iron ore

steel

limestone

hot air

slag

iron

coke

waste gases

Choose substances from the box to complete the following sentences.

[2]

Iron oxide, coke and are the materials fed in at the **top** of the furnace.

The furnace is heated by burning the coke in

(b) Balance the symbol equation that represents the main reaction taking place inside the furnace.

[1]



- (c) Some of the impurities are removed from the furnace as slag. Slag contains calcium silicate which has the chemical formula CaSiO_3 .

Tick (✓) the box next to the calculation used to find the correct relative formula mass of calcium silicate, CaSiO_3 . [1]

$$A_r(\text{Ca}) = 40$$

$$A_r(\text{Si}) = 28$$

$$A_r(\text{O}) = 16$$

$$40 + 28 + 16$$

☐

$$40 + 40 + 40 + 28 + 28 + 28 + 16 + 16 + 16$$

☐

$$40 + 28 + 16 + 16 + 16$$

☐

$$40 + 16 + 16 + 16$$

☐

$$40 + 28 + 28 + 28 + 16 + 16 + 16$$

☐

- (d) 820 tonnes of iron was extracted from 1 750 tonnes of iron ore. Calculate the percentage of iron in the ore. Give your answer to **one** decimal place. [2]

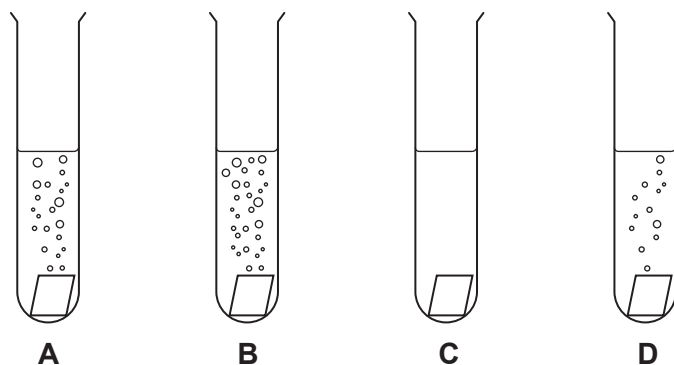
Percentage = %



3. Nathan and Simon were investigating the reactivity of metals.

- (a) In one experiment, they investigated the effect of adding dilute hydrochloric acid to four different metals, **A**, **B**, **C** and **D**.

The diagram shows the observations made.



- (i) They used the same **volume** of acid in each tube. Give **two other** ways they controlled the dilute hydrochloric acid to ensure a fair test. [2]

1.
2.

- (ii) Use the observations to list the metals in order of their reactivity. Give the reason for your choice. [2]

Most reactive

.....

.....

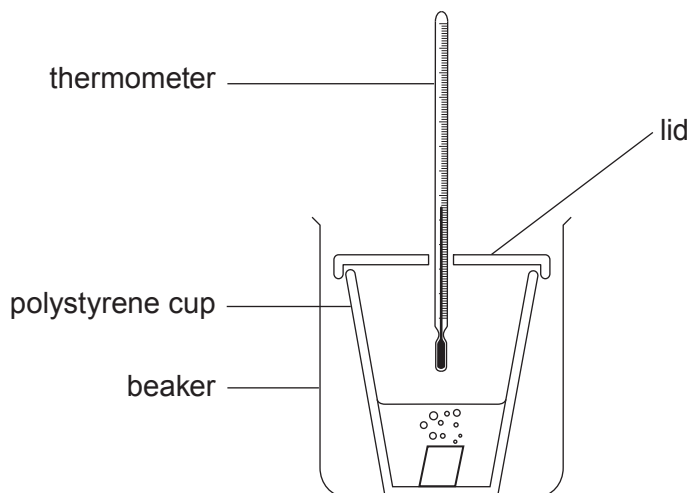
Least reactive

Reason

.....



- (b) In a second experiment, Nathan and Simon used the following apparatus to investigate the temperature rise when metals **A**, **B**, **C** and **D** were added to hydrochloric acid.



- (i) Identify the error in the way the apparatus is set up and state how this would affect the results collected. [2]

Error

Effect

.....



(ii) Once they had set up their apparatus correctly, they obtained the following results.

Metal	Temperature rise (°C)			
	Test 1	Test 2	Test 3	Mean
A	22	21		22
B	40	38	36	38
C	0	0	0	0
D	15	9	13	14

I. Identify the anomalous result from metals **B**, **C** and **D**. [1]

Metal Test

II. Calculate the missing result for metal **A**. [1]

Temperature rise for test 3 = °C

III. Zinc is more reactive than metals **C** and **D** but less reactive than metals **A** and **B**. Use this information to predict the temperature rise that would be expected for zinc. [1]

Temperature rise for zinc °C



- (c) The reaction of a metal with an acid is summarised by the following equation.



- (i) Complete the equation for the reaction between magnesium and hydrochloric acid by

- giving the formula of the salt magnesium chloride,
- balancing the equation.

[2]



- (ii) Give the name of the salt formed when magnesium reacts with nitric acid.

[1]

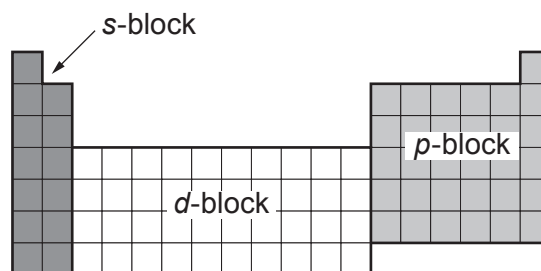
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4. The Transition Metals

The elements in the Periodic Table can be divided into three blocks: the *s*-block, the *d*-block and the *p*-block.

The *s*-block and *d*-block contain only metallic elements. The *p*-block contains both metallic and non-metallic elements.



The transition metals are found within the *d*-block and are the metallic elements that serve as a bridge or 'transition' between the two sides of the table. They include the elements iron, silver, gold, vanadium and titanium.

The transition metals have similar properties to each other. These include high melting point, high density, good conductivity and malleability. The transition metals also have unique properties, making them different from the main group metals, including the ability to form compounds with different colours and different oxidation states.

Table 1 gives the properties of some transition metals and some Group 1 metals.

Element	Melting point (°C)	Boiling point (°C)	Oxidation states	Colours of compounds formed
iron	1536	2861	+2, +3	green and brown
lithium	180	1342	+1	white
potassium	63	760	+1	white
sodium	98	883	+1	white
titanium	1660	3287	+3, +4	violet and white
vanadium	1910	3407	+3, +4, +5	green, blue and yellow

Table 1



- (a) Tick (✓) the statement that **best** describes where metals are positioned within the Periodic Table. [1]

the s-block, *p*-block and *d*-block

☐

the s-block and *p*-block only

☐

the s-block only

☐

the *d*-block and *p*-block only

☐

the *p*-block only

☐

- (b) Name the element in **Table 1** that is a liquid over the biggest temperature range. [1]

.....

- (c) Tick (✓) the **two** statements that correctly describe the oxidation states of the metals listed in **Table 1**. [2]

the Group 1 metals and transition metals all have a +1 oxidation state

☐

the transition metals all have a +3 oxidation state

☐

the Group 1 metals all have a +1 oxidation state

☐

iron and lithium have the same oxidation states

☐

the Group 1 metals and transition metals all have a +4 oxidation state

☐

(d) Consider the following statement.

‘It is possible to tell the Group 1 and transition metals apart by the colour of the compounds they form.’

Use the information in **Table 1** to give **one** reason to agree and **one** reason to disagree with this statement. [2]

Reason to agree

.....

.....

Reason to disagree

.....

.....



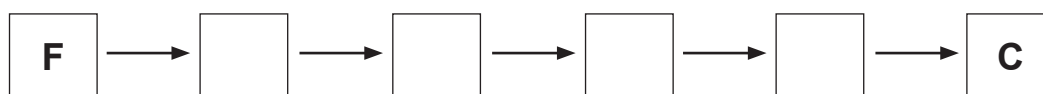
5. (a) The first stage in the preparation of a salt from an acid and an alkali often involves carrying out a titration.



A-F below are the steps of the method that can be used to prepare a salt from an acid and an alkali. However, these steps are **not** in the correct order.

- A** Repeat without the indicator using the same volumes of alkali and acid
- B** Add 2-3 drops of indicator
- C** Allow the solution to cool and crystallise naturally
- D** Place the salt solution into an evaporating dish and heat until half the volume of solution remains
- E** Add the acid to the alkali until the indicator changes colour
- F** Measure 25.0 cm^3 of alkali into a conical flask

- (i) Place the steps of the method into the correct order. The first and last steps have already been added. [2]



- (ii) Give the **letter** of the step where error is **most** likely to occur. Give the reason for your choice. [2]

Step

Reason

.....



- (b) The reaction between an acid and an alkali produces a salt and water. The following equation shows how potassium sulfate is produced from an acid and an alkali.

potassium hydroxide +[?]..... → potassium sulfate + water

- (i) Give the name of the missing acid.

[1]

.....

- (ii) Give the chemical formula of potassium sulfate.

[1]

.....



Examiner
only

6. The table shows some properties of aluminium, copper and titanium.

Metal	Melting point	Electrical conductivity	Thermal conductivity	Density	Resistance to corrosion
aluminium	high	good	good	low	good
copper	high	good	good	high	good
titanium	high	good	good	low	good

Describe how **these** properties are linked to the uses of aluminium, copper and titanium.

You should **not** give more than two uses for any one metal. [6 QER]

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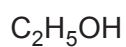
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.....

6



7. (a) The molecular formulae of five carbon compounds, **A**, **B**, **C**, **D** and **E** are shown below.

**A****B****C****D****E**

- (i) Alkanes are compounds with the general formula $\text{C}_n\text{H}_{2n+2}$.

Give the **letters** of the **two** compounds that are alkanes.

[1]

..... and

- (ii) Give the **name** for compound **A**.

[1]

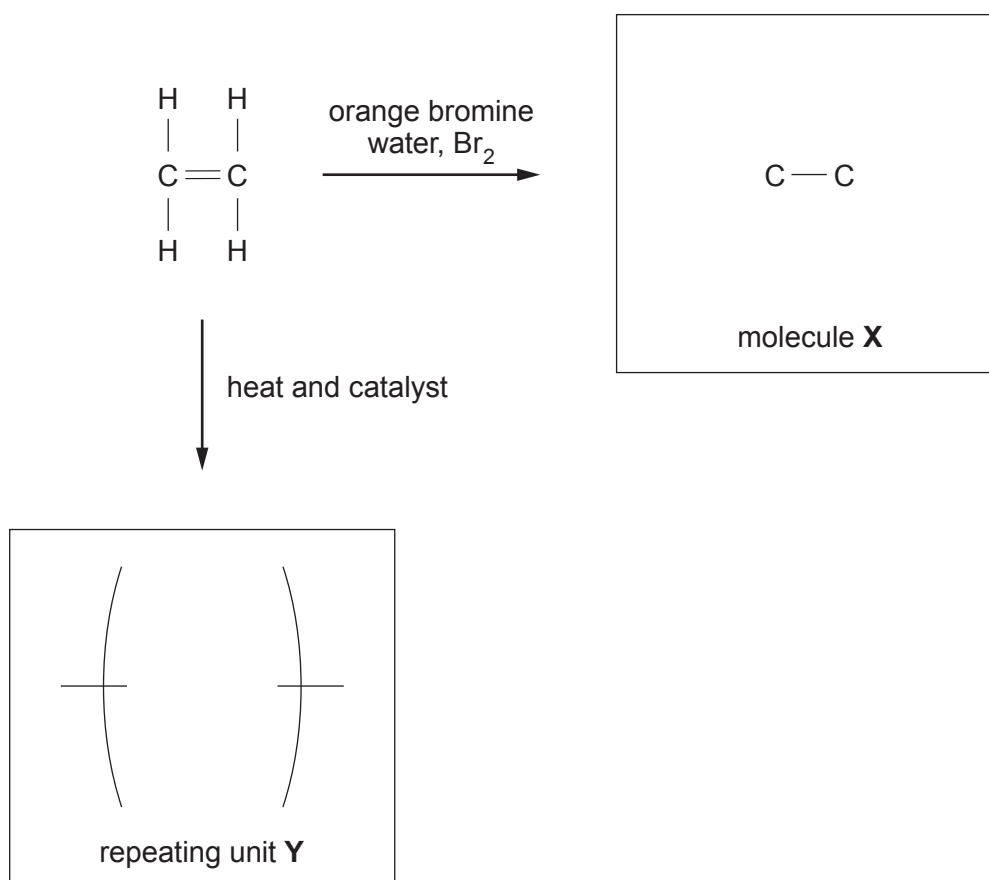
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- (iii) Draw the structural formula of compound **E**.

[1]



(b) The following diagram shows two reactions of ethene.



(i) **Complete the structural formulae** for molecule **X** and repeating unit **Y**. [2]

(ii) Describe the change that would be seen in the orange bromine water when molecule **X** is formed. [1]

.....

(iii) Name the type of reaction that takes place to form repeating unit **Y**. [1]

.....



(c) Paper and plastic are both used to make disposable shopping bags.



The table compares the percentages of disposable paper and plastic bags used by consumers in a Welsh town between 1988 and 2018.

Type of bag	Percentage of disposable bags used (%)						
	1988	1993	1998	2003	2008	2013	2018
paper	73	63	31	12	8	13	21
plastic	27	37	69	88	92	87	79

(i) In 2003, the consumers of the town used a total of 1.25 million disposable bags.

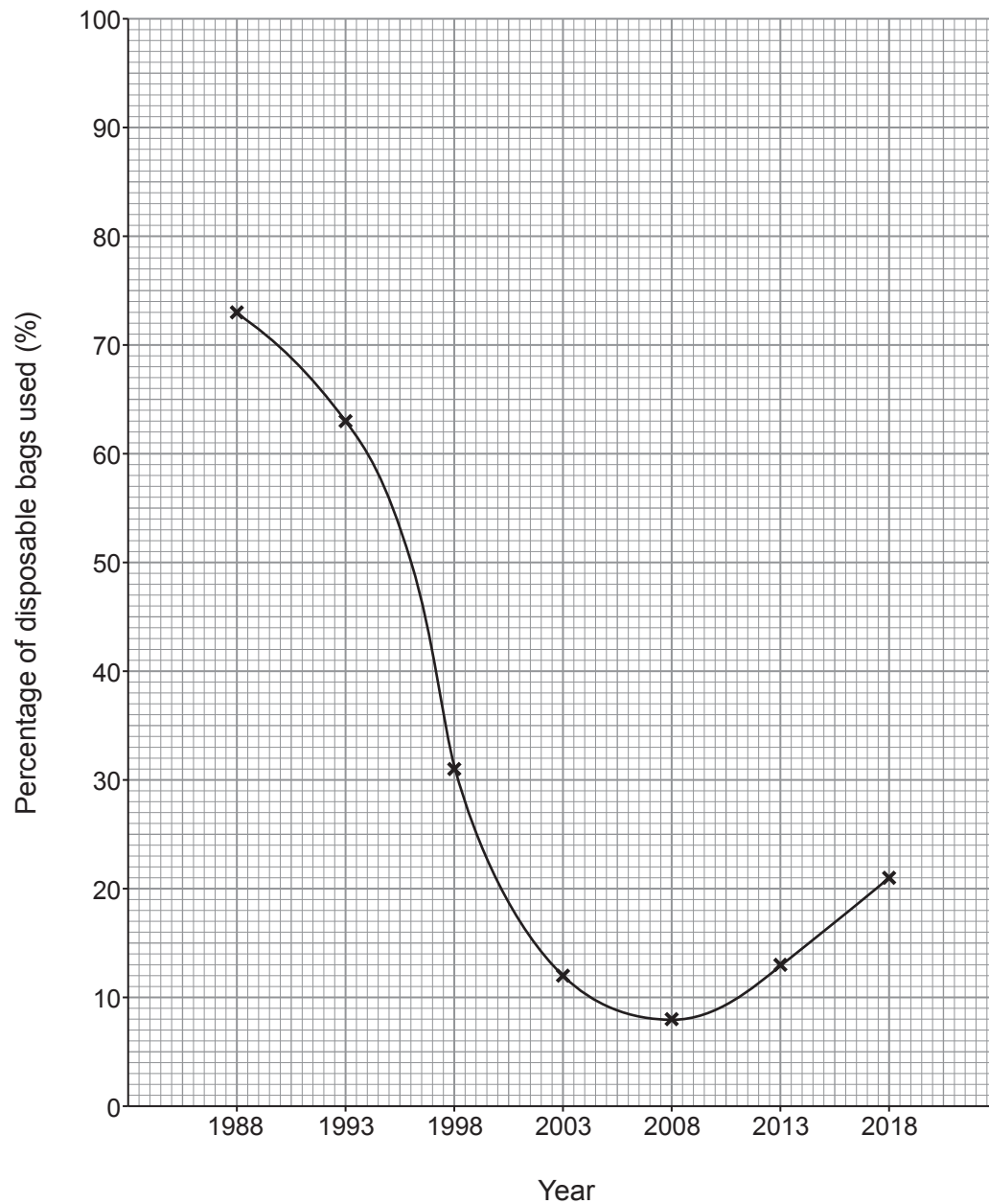
Use the information in the table to calculate the number of **paper** bags used. [2]

Number of paper bags = million



- (ii) The graph below shows the percentage of **paper** bags used between 1988 and 2018.

Plot on the **same grid** the percentage of **plastic** bags used between 1988 and 2018. Draw a suitable line. [3]



Examiner
only

(iii) Use the graph to answer the following questions.

I. Give the year when **equal** numbers of plastic bags and paper bags were used. [1]

Year

II. Circle the **simplest** ratio that compares the use of paper bags to plastic bags in 2000. [1]

2 : 8 1 : 4 4 : 1 8 : 2 20 : 80

(iv) Give **one** reason that explains the change seen in plastic bag usage in Wales between 2008 and 2018. [1]

.....

.....

15

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0



28

1	H
Hydrogen	1

7	Li	9	Be	11	B	12	C	14	N	16	O	19	F	20	Ne
Lithium	3	Beryllium	4	Boron	5	Carbon	6	Nitrogen	7	Oxygen	8	Fluorine	9	Neon	10
23	Na	24	Mg	27	Al	28	Si	31	P	32	S	35.5	Cl	40	Ar
Sodium	11	Magnesium	12	Aluminium	13	Silicon	14	Phosphorus	15	Sulfur	16	Chlorine	17	Argon	18
39	K	40	Ca	45	Sc	48	Ti	51	V	52	Cr	55	Mn	56	Fe
Potassium	19	Calcium	20	Scandium	21	Titanium	22	Vanadium	23	Chromium	24	Manganese	25	Iron	26
86	Rb	88	Sr	89	Y	91	Zr	93	Nb	96	Mo	99	Tc	101	Ru
Rubidium	37	Strontium	38	Yttrium	39	Zirconium	40	Niobium	41	Molybdenum	42	Technetium	43	Ruthenium	44
133	Cs	137	Ba	139	La	179	Hf	181	Ta	184	W	186	Re	190	Os
Caesium	55	Barium	56	Lanthanum	57	Hafnium	72	Tantalum	73	Tungsten	74	Rhenium	75	Osmium	76
223	Fr	226	Ra	227	Ac	204	Tl	207	Pb	209	Bi	210	Po	210	At
Francium	87	Radium	88	Actinium	89	Thallium	81	Lead	82	Bismuth	83	Polonium	84	Astatine	85
65	Zn	63.5	Cu	59	Ni	59	Co	58	Ni	63.5	Cu	65	Zn	70	Ga
Zinc	30	Copper	29	Cobalt	27	Nickel	28	Gallium	31	Germanium	32	Arsenic	33	Se	34
112	Cd	108	Ag	106	Pd	106	Rh	105	In	115	Sn	122	Sb	127	I
Cadmium	48	Silver	47	Rhodium	45	Palladium	46	Indium	49	Tin	50	Antimony	51	Tellurium	52
201	Hg	197	Au	195	Pt	195	Ir	201	Hg	204	Tl	207	Pb	210	Po
Mercury	80	Gold	79	Platinum	78	Iridium	77	Mercury	80	Thallium	81	Lead	82	Polonium	84
227	Fr	226	Ra	227	Ac	204	Tl	207	Pb	209	Bi	210	Po	210	At
Francium	87	Radium	88	Actinium	89	Thallium	81	Lead	82	Bismuth	83	Polonium	84	Astatine	85

Key

Ar	relative atomic mass
Symbol	
Name	
Z	atomic number